Leveraging Open Source to Build Digital Public Infrastructure that is Inclusive, Transparent, and Scalable

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Executive Summary

Digital Public Infrastructure is already reshaping how societies function. By integrating identity, payments, and data exchange systems, countries can expand access to essential services, increase government efficiency, and create fertile ground for innovation. Yet technology alone does not guarantee success. Effective DPI requires governance, trust, and financial sustainability, and above all, it requires openness, the only way to ensure transparency, adaptability, and long-term resilience. Without transparency, adaptability, and interoperability, infrastructure risks becoming exclusionary, brittle, or dependent on a handful of private vendors.

Open source offers a proven path forward. Platforms such as MOSIP (identity) and Mojaloop (payments) demonstrate how open design can support scalability, trust, and innovation across diverse contexts. Philanthropic organizations, including the Gates Foundation, have helped de-risk adoption by funding and stewarding these platforms and helping countries implement a DPI as a global public good.

The central argument of this article is that DPI must be treated as a strategic asset and designed with openness as the default. Governments, multilateral

institutions, and the private sector each have roles to play in ensuring these systems remain inclusive, sustainable, and resilient over time.

Defining the DPI Stack

To understand the impact and importance of DPI, we first need to define what the stack encompasses. DPI is not a single technology but a layered architecture that relies on three interdependent components: digital identity, digital payments, and data exchange frameworks. Each component reinforces the others, creating a foundation for secure, scalable, and inclusive digital services.

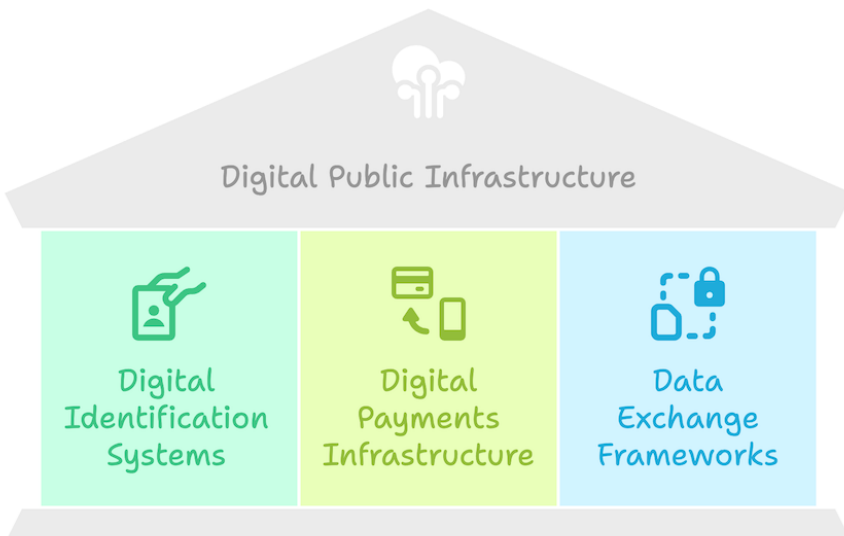

Figure 1:

1. **Digital Identification Systems:** Digital identity systems allow individuals to prove who they are in a secure and verifiable manner. These systems form the gateway to public and private services, enabling citizens to open bank accounts, access healthcare, and receive social benefits. The strength and reliability of digital IDs are critical because they underpin all subsequent digital interactions.

2. **Digital Payments Infrastructure:** Payments infrastructure allows money to move quickly, safely, and efficiently between individuals, businesses, and governments. Its interoperability is essential: without standard protocols and rails, payments remain fragmented and exclusionary. Digital payments amplify the impact of identity systems by connecting users to financial and commercial ecosystems.

3. **Data Exchange Frameworks:** Data exchange frameworks govern how

information is shared, reused, and protected across platforms and sectors. These frameworks enable integration between health, education, taxation, and other systems, unlocking efficiencies and insights while maintaining security and privacy.

The value of DPI lies in how its layers reinforce one another: identity enables payments, payments expand participation, and data exchange integrates services. This compounding effect is only possible when systems are designed to interoperate, and that's best achieved through open standards and open source.

DPI as a Catalyst for Inclusion, Efficiency, and Growth

DPI is a transformative tool that drives inclusion, efficiency, and innovation, and its effects compound: inclusion improves efficiency, efficiency enables innovation, and innovation accelerates growth. Its adoption has far-reaching implications, not only for national economies but also for global competitiveness and social equity.

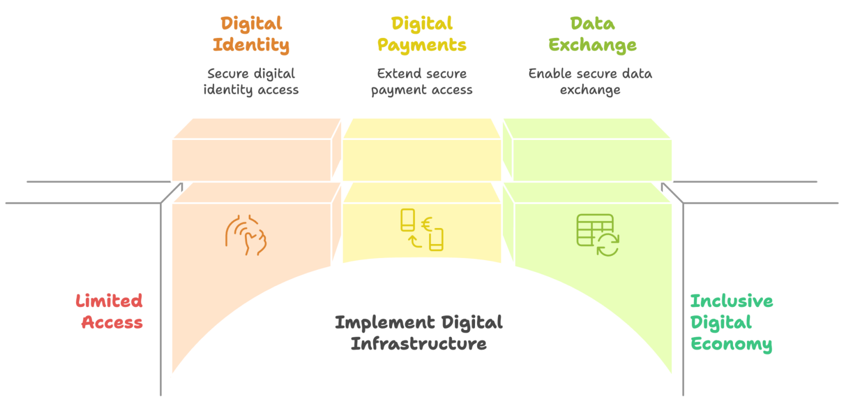

Figure 2:

- **Economic Inclusion:** One of DPI's most profound impacts is economic or financial inclusion. By providing citizens with secure digital identities and access to payment infrastructure, governments can bring previously unbanked populations into the formal economy. This access to secure digital identities and access to payment creates opportunities for savings, credit, and participation in markets that were previously inaccessible.
- **Government Efficiency:** DPI also enables governments to deliver services more effectively. By digitizing identification and payment processes, governments can reduce fraud, increase transparency, and streamline service delivery. Efficient DPI reduces administrative overhead and ensures that resources reach intended beneficiaries.
- **Innovation and Growth:** Beyond inclusion and efficiency, DPI creates

a platform for innovation. Once digital rails are in place, entrepreneurs, startups, and established companies can develop new applications and services that leverage identity, payments, and data exchange. This wave of innovation catalyzes economic growth and strengthens national competitiveness.

While DPI's benefits are clear, achieving them requires coordinated efforts among multiple actors. Crucially, these benefits are cumulative: inclusion improves efficiency, efficiency enables innovation, and innovation accelerates growth. But the full effect only materializes when DPI is open, interoperable, and trusted,

The Role of Global Actors

DPI does not develop in isolation, and its ecosystem depends on the collaboration among multiple stakeholders, each bringing unique strengths. These stakeholders include governments, multilateral organizations, open source communities, and philanthropic foundations, all of which play pivotal roles in shaping and sustaining digital infrastructure.

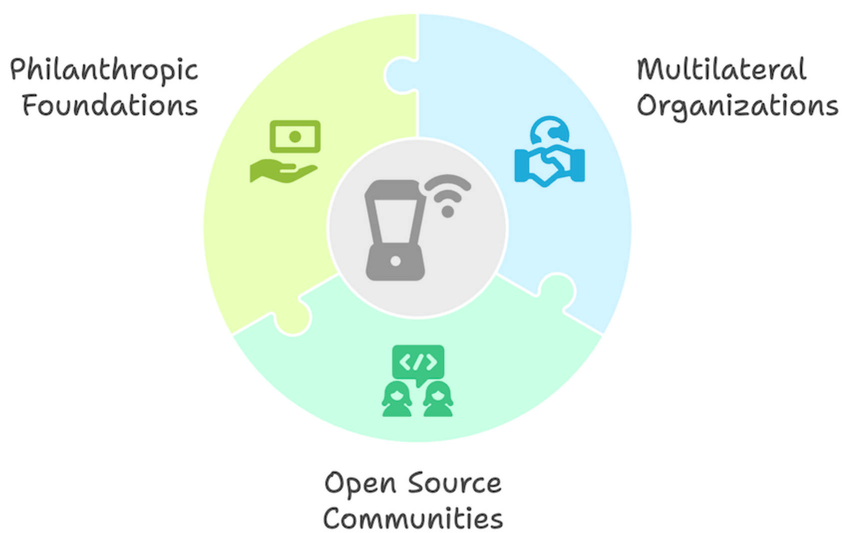

Figure 3:

- **Multilateral Organizations:** Multilateral institutions provide policy guidance, financing, and standards for interoperability. They help coordinate efforts across countries, ensuring that digital systems are aligned and scalable.
- **Open Source Communities:** Open source communities create plat-

forms that are transparent, adaptable, and cost-effective. Projects like MOSIP (identity), Mojaloop (payments), and DHIS2 (health data) exemplify collaborative approaches where governments and developers can contribute, audit, and adapt infrastructure to meet local needs.

- **Philanthropic Foundations:** Foundations, such as the Gates Foundation, have been instrumental in supporting open source DPI initiatives. Through funding, convening stakeholders, and providing technical guidance, philanthropy reduces risks for early adopters and helps create infrastructure that benefits countries and citizens without locking them into proprietary solutions.

These actors form an ecosystem: multilateral legitimacy, open source adaptability, and philanthropic risk-sharing reinforce one another to create sustainable DPI.

Key Challenges

Despite its promise, DPI is not without risks and obstacles. Some of the key challenges, such as adoption, capacity, and governance, can undermine its effectiveness if not addressed proactively.

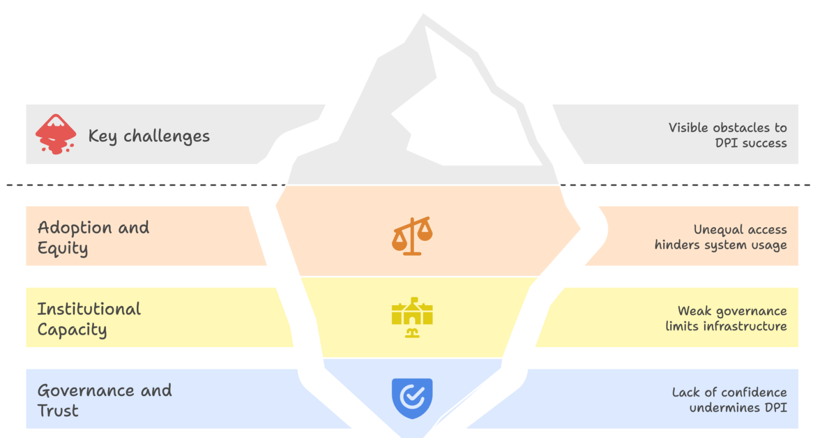

Figure 4:

- **Adoption and Equity:** Even robust systems can fail if citizens are unable or unwilling to use them. Barriers such as limited literacy, device access, and affordability can reinforce inequality unless inclusion is designed into the system from the start. Open source helps by enabling local adaptation and lowering costs.
- **Institutional Capacity:** Digital infrastructure requires strong governance, legal frameworks, and sustainable financing. Without them, coun-

tries risk becoming dependent on external vendors or experiencing uneven implementation. Open source platforms help build local skills and reduce reliance on proprietary providers.

- **Governance and Trust:** Trust is the foundation of DPI. Citizens must have confidence that their data is secure and that systems are accountable. Weak governance can lead to mistrust, low adoption, and even systemic failure. Open code and open governance processes allow for independent scrutiny, reducing risks of misuse and strengthening legitimacy.

These risks can be mitigated by openness: local adaptation lowers barriers to use, open communities build institutional capacity, and transparency strengthens trust.

Open Source as the Backbone of Digital Public Infrastructure

For infrastructure to function as a public good, its foundations must be transparent and adaptable. Proprietary systems create dependency, limit interoperability, and undermine trust. Open source, by contrast, allows governments and communities to inspect, audit, and evolve core systems.

DPI and the Case for Openness

Open source platforms allow governments and stakeholders to audit, adapt, and improve infrastructure without relying on proprietary vendors. Initiatives like MOSIP and Mojaloop demonstrate how open source supports scalability, trust, and innovation across contexts.

Advantages of Open Source in DPI

Open source strengthens DPI by promoting transparency, reducing costs, enabling local customization, ensuring interoperability, and fostering ecosystem development. These advantages collectively make digital infrastructure more resilient, inclusive, and sustainable.

> Embedding openness at the core of DPI sets the stage for long-term success and prepares the ecosystem for the next phase of development.

Conclusion

DPI is the backbone of modern digital economies and public services. Its long-term success depends on technology, as well as governance, trust, and strategic adoption of open source. Leaders must treat DPI as a public good, with open source as its foundation, investing in capacity, transparency, and inclusivity. Open source is essential to achieving these objectives, providing the flexibility and accountability necessary to scale infrastructure responsibly.

- **Governments** should embed DPI into national development strategies with open design as the default.
- **Philanthropies and multilateral agencies** should continue supporting open platforms as global public goods.
- **Private sector partners** should innovate on top of open infrastructure while respecting its shared nature.

Countries worldwide are already implementing DPI, with varying models and levels of openness. The question is not whether DPI will be built, but whether it will be built openly. Closed systems concentrate power and fragility, while open systems distribute trust and resilience.

About the Author

Dr. Ibrahim Haddad is Head of Infotainment at Volvo Cars. He previously served as Vice President of Strategic Programs at the Linux Foundation, where he was Executive Director of LF AI & Data and the founding Executive Director of the PyTorch Foundation. In these roles, he led global open source AI initiatives, scaling ecosystems, strengthening governance, and driving cross-industry collaboration among leading technology companies. Earlier, Dr. Haddad served as Vice President of R&D and Distinguished Engineer at Samsung Research. Throughout his career, he has held technical and leadership positions at Ericsson Research, Motorola, Palm, HP, and the Open Source Development Labs. He earned a Ph.D. in Computer Science from Concordia University (Montreal, Canada) and has lived and worked across North America, Europe, and Asia. (LinkedIn, Website)